

DRL-Based Single-Input Pinning Synchronization of Hopfield Chaotic Neural Networks with an Application to Image Encryption

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ABSTRACT

We develop a deep reinforcement learning (DRL)-based single-input pinning synchronization control framework for continuous-time Hopfield chaotic neural networks and explore its application to image encryption. In the proposed framework, only one scalar control input is injected into a selected node of the response system, and proximal policy optimization (PPO) is adopted as a representative DRL algorithm to learn the constrained nonlinear control policy. To support pinning-node selection, a multi-index local analysis is performed, and the drive-response synchronization process is formulated as a continuous-state and continuous-action reinforcement learning problem. Numerical results show that the local node ranking is generally consistent with the closed-loop synchronization behavior. Fixed-gain feedback, LQR, and sliding-mode control are included as traditional baselines. Under the more challenging Node 2 channel and the largest tested parameter mismatch, PPO obtains an MAE of 0.1703, compared with 0.2330 for LQR and 0.2223 for sliding-mode control; after a global impulse, its post-disturbance mean error is 2.0800, compared with 2.5264 and 2.7691, respectively, although the model-based baselines require less control energy. A post-training trajectory assessment further yields negative largest conditional Lyapunov exponents for all evaluated Node 2 and Node 3 closed-loop trajectories, providing local numerical evidence of average transverse error decay. Based on the synchronized chaotic trajectories, an image encryption and decryption scheme is constructed using Logistic-map diffusion and dynamic DNA coding. Statistical analysis shows that the encrypted images exhibit reduced statistical redundancy. These results support the feasibility of DRL-based single-input pinning control for practical low-residual synchronization under constrained actuation and limited observation.

1. Introduction

Chaotic systems have been widely studied in nonlinear science [1], complex networks [2], and neural dynamics [3] because of their sensitive dependence on initial conditions, a property in which small differences in initial states may grow rapidly and lead to markedly different trajectories over time. This sensitivity makes chaotic systems difficult to predict and control, and also raises the fundamental problem of whether chaotic motions can be coordinated or synchronized through appropriate coupling or control. In 1990, Pecora and Carroll demonstrated drive-response synchronization in chaotic systems [4], showing that two chaotic systems can evolve coherently under suitable conditions. Since then, the analysis, control, and synchronization of chaotic systems have attracted extensive attention, with applications in secure communication [5–7], image encryption [8, 9], and coordinated control of complex systems [10, 11].

For chaotic systems, synchronization refers to the process by which two or more systems evolve toward a consistent dynamical relationship through appropriate coupling or control mechanisms. Different synchronization forms have been widely investigated, including complete synchronization [12], generalized synchronization [13], phase synchronization [14], and lag synchronization [15]. To realize these synchronization objectives and control chaotic

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behaviors, a variety of classical control methods have been developed [16, 17], such as linear feedback control [18], adaptive control [19], sliding-mode control [20–22], impulsive control [23], and pinning control [24].

Although these classical methods have provided effective tools for stabilizing and synchronizing chaotic systems, they usually rely on explicit system models, stability analysis, and carefully designed feedback laws. For complex nonlinear systems with uncertain dynamics, limited observation information, or difficult-to-model behaviors, such requirements may restrict their applicability. To reduce the dependence on analytical controller design, the deep reinforcement learning (DRL) has been increasingly explored as a model-free data-driven control approach for complex dynamical systems [25–27]. Through interaction with the environment, DRL learns nonlinear control policies directly from state observations and reward feedback, making it suitable for continuous control and decision-making tasks. Existing studies have introduced learning-based methods into chaos control and synchronization tasks, showing that DRL controllers can achieve effective control or synchronization in several typical chaotic systems [28–34]. Some recent works [35, 36] have further considered reduced control inputs, noisy environments, or incomplete state information, suggesting that learning-based controllers may retain certain adaptability under non-ideal conditions.

As a representative neural dynamical model, the Hopfield neural network has also been widely used to study nonlinear dynamics and chaos. In the classical graded-response Hopfield model, suitable symmetric-connection conditions usually lead the network state to converge to equilibrium points [37]. However, the modified Hopfield-type neural networks can exhibit rich nonlinear dynamics, including bifurcation, multistability, and chaotic attractors [38–40]. Existing studies on Hopfield-type chaotic neural networks have mainly focused on dynamical analysis and model-based control or synchronization, such as fractional-order dynamics, chaos control, weak synchronization, and impulsive or intermittent synchronization [41–44]. Recent studies have further investigated neural-network synchronization under event-triggered control, pinning control, delayed dynamics, and image-encryption-oriented applications [45–47]. These studies provide useful foundations for understanding and controlling Hopfield chaotic dynamics, especially from the perspectives of dynamical characterization, model-based synchronization, and secure-information applications.

Despite recent progress in both Hopfield chaotic dynamics and DRL-based chaos control, the synchronization of continuous-time Hopfield chaotic neural networks under constrained pinning control has received less attention. Many existing DRL-based chaotic synchronization studies have mainly focused on typical low-dimensional chaotic systems, such as the Lorenz system, Rössler system, and Chua circuit [31, 32], or have adopted relatively flexible multi-variable control structures. In contrast, a single-input pinning architecture uses only one actuator channel and injects one scalar signal into a selected response node. This restriction is relevant when actuator count, sensing, communication, or implementation complexity limits the available control channels; it also makes the pinning-node choice critical to the performance of the learned controller. Moreover, practical control systems are often subject to actuator limitations, model uncertainty, external perturbations, stochastic measurement or process noise, and limited sensing information [36, 42, 43]. In this context, single-input pinning control can be interpreted as a constrained-actuation scenario, where only one control channel is available for synchronization. Therefore, it is necessary to evaluate not only nominal synchronization performance but also the robustness and adaptability of the learned controller under non-ideal operating conditions.

In addition to synchronization control itself, chaotic synchronization also has potential value in secure image encryption. Owing to their sensitivity to initial conditions and pseudo-random dynamical behavior, chaotic systems have been widely used to construct key streams, permutation–diffusion mechanisms, and DNA-coding-based encryption procedures [8, 48–51]. In a drive–response synchronization framework, the synchronized chaotic trajectories can provide consistent key streams for encryption and decryption, while their irregular dynamics help reduce the statistical redundancy of the encrypted image. Therefore, image encryption offers a practical application scenario for examining whether the learned synchronization controller can generate usable chaotic key streams beyond the synchronization task itself.

Motivated by these considerations, this paper investigates the single-input pinning synchronization problem for a continuous-time Hopfield chaotic neural network and further explores its application to image encryption. A DRL-based data-driven framework is developed, in which proximal policy optimization (PPO) is adopted to learn a constrained nonlinear control policy. The proposed method integrates local pinning-node analysis with reinforcement-learning-based control, allowing the node-selection result and the learned policy to be evaluated within a unified drive–response synchronization setting. The learned controller is further tested under nominal synchronization, parameter mismatch, global impulse disturbance, Gaussian noise perturbation, and partial-observation settings. Finally, the

synchronized chaotic trajectories are used to construct an image encryption and decryption scheme based on Logistic-map diffusion and dynamic DNA coding. The main contributions of this paper are summarized as follows:

- A DRL-based synchronization control framework is developed for two continuous-time Hopfield chaotic neural networks, where the nonlinear feedback policy is learned from interaction data rather than prescribed analytically.
- A single-input pinning control strategy is formulated, in which only one scalar control signal is applied to a selected node of the response system, representing a constrained-actuation setting with a single available control channel.
- A multi-index local pinning-node ranking is constructed by combining the maximum conditional Lyapunov exponent, Riccati-based cost, feedback gain norm, and local control propagation index.
- The relationship between pinning-node choice and observation demand is examined under full and reduced observation spaces, and the empirical adaptability of the PPO-implemented controller is evaluated under parameter mismatch, global impulse disturbance, and Gaussian noise perturbation. Traditional baselines, a matched-budget SAC comparison, and all nonempty actuator subsets quantify the associated error–effort and actuation tradeoffs.
- A chaos-based image encryption scheme is developed using synchronized Hopfield trajectories, Logistic-map diffusion, and dynamic DNA coding. This application extends the use of DRL-based chaotic synchronization from control-oriented tasks to secure image protection.

The remainder of this paper is organized as follows. Section 2 introduces the continuous-time Hopfield chaotic neural network and formulates the single-input pinning synchronization problem. Section 3 presents the local pinning-node selection analysis and the DRL-based control framework implemented by PPO. Section 4 reports the experimental setup, results, and discussion. Section 5 presents the application in image encryption. Finally, Section 6 concludes this paper and outlines possible future work.

2. Preliminaries

Hopfield's continuous neural network model proposed in 1984 [37] can be written as

$$C_i \frac{du_i}{dt} = \sum_{j=1}^n T_{ij} V_j - \frac{u_i}{R_i} + I_i, \quad (1)$$

with the static nonlinear input–output mapping $V_i = g_i(u_i)$. Here, u_i denotes the internal state variable of the i th neuron, V_i is the neuron output, C_i and R_i represent the equivalent capacitance and resistance, respectively, T_{ij} denotes the synaptic connection weight, and I_i is the external bias input.

In Hopfield's original study, the matrix T was assumed to be symmetric so as to guarantee the monotonic decrease of the system energy function in a mathematical sense. However, such a setting leads the network to eventually converge to a static equilibrium point and thus prevents the emergence of chaotic behavior. To focus on the nonlinear bifurcation and chaotic attractor characteristics induced by the complex internal topological coupling of the neural network, the original equations are often nondimensionalized in the control literature [38, 41].

Introducing the parameter substitutions $a_i = \frac{1}{R_i C_i}$, $w_{ij} = \frac{T_{ij}}{C_i}$, and $b_i = \frac{I_i}{C_i}$, and assuming that all neurons share the same leakage coefficient ($a_i = 1$), neglecting the constant bias term ($\mathbf{b} = \mathbf{0}$), and choosing $\tanh(\cdot)$ to characterize the saturating nonlinearity of the neuron output, one obtains the continuous-time Hopfield-type neural network model used in the subsequent analysis:

$$\dot{\mathbf{x}}(t) = -\mathbf{x}(t) + \mathbf{W} \tanh(\mathbf{x}(t)), \quad (2)$$

where $\mathbf{x}(t) = [x_1(t), x_2(t), x_3(t)]^T$ is the state vector of the system (2). Following the three-neuron Hopfield-type chaotic model studied in [41], in this paper, the network weight matrix is specified as

$$\mathbf{W} = \begin{bmatrix} -1.4 & 1.2 & -7.0 \\ 1.1 & 0.0 & 2.8 \\ 0.8 & -2.0 & 4.0 \end{bmatrix}. \quad (3)$$

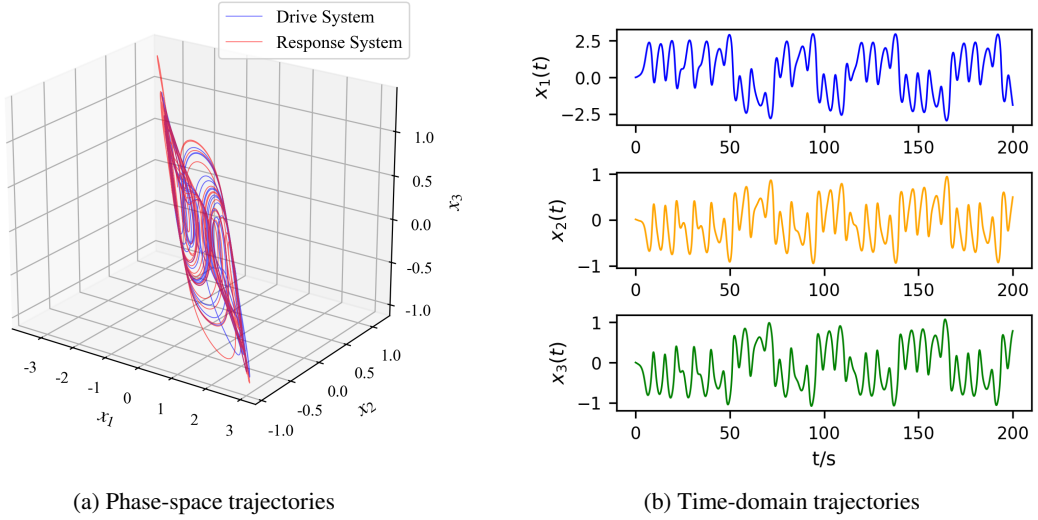


Figure 1: Uncontrolled phase-space trajectories of the drive and response Hopfield chaotic neural networks and time-domain state trajectories of the continuous-time Hopfield-type chaotic neural network.

To investigate drive–response synchronization, the Hopfield-type chaotic neural network described by (2) is taken as the drive system. The response system is constructed by introducing a single-input pinning controller into one selected node, and its dynamics can be generally written as:

$$\dot{\mathbf{y}}(t) = -\mathbf{y}(t) + \mathbf{W}_r \tanh(\mathbf{y}(t)) + \mathbf{B}u(t), \quad (4)$$

where $\mathbf{y}(t) \in \mathbb{R}^3$ is the state of the response system; $\mathbf{W}_r$ is the weight matrix of the response system, which is allowed to have slight hardware-induced parameter mismatch with respect to $\mathbf{W}$; $u(t) \in \mathbb{R}$ is the scalar control input; and $\mathbf{B} \in \mathbb{R}^{3 \times 1}$ is the pinning-node selection vector, which determines the specific physical node into which the control signal is injected. Only one of its three components is equal to 1, while the others are 0.

For this parameter setting, the initial condition $\mathbf{x}(0) = (0, 0.01, 0)^T$, which has been used as a representative initial state in previous studies on Hopfield chaotic neural networks [41], is adopted to illustrate the basic chaotic behavior of the system. The fourth-order Runge–Kutta (RK4) simulation results show that the system exhibits chaotic behavior. To further illustrate the uncontrolled drive–response behavior before applying the proposed pinning controller, two identical Hopfield chaotic neural networks are simulated with the same system parameters but different initial states sampled from Uniform($[-1, 1]^3$). As shown in Fig. 1(a), the two trajectories evolve around the same double-scroll chaotic attractor but do not naturally synchronize.

The time-domain trajectories of $x_1(t)$, $x_2(t)$, and $x_3(t)$ obtained from the representative initial condition $\mathbf{x}(0) = (0, 0.01, 0)^T$ are shown in Fig. 1(b). These trajectories exhibit non-periodic oscillatory behaviors, further indicating the chaotic dynamical characteristics of the continuous-time Hopfield neural network under the current parameter setting.

Define the synchronization error as $\mathbf{e}(t) = \mathbf{y}(t) - \mathbf{x}(t)$. Exact synchronization corresponds to asymptotic stability of the zero solution of the error system [52], whose dynamics are

$$\dot{\mathbf{e}}(t) = -\mathbf{e}(t) + \mathbf{W}_r \tanh(\mathbf{y}(t)) - \mathbf{W} \tanh(\mathbf{x}(t)) + \mathbf{B}u(t). \quad (5)$$

The following definition mainly corresponds to the nominal synchronization control problem.

Definition 1. For the drive system (2) and the response system with pinning control (4), if under the action of the control law $u(t)$ one has $\lim_{t \rightarrow \infty} \|\mathbf{e}(t)\| = 0$, then the systems are said to achieve asymptotic synchronization.

This definition provides the ideal exact-synchronization reference. Because the learned policy does not enforce invariance of $\mathbf{e} = \mathbf{0}$ by construction, its achieved behavior is evaluated as practical synchronization with a small residual error, together with the post-training local assessment introduced in Section 3.3.

3. PPO-Based Single-Input Pinning Control Method

Deriving an effective control law $u(t)$ for the nonlinear underactuated error system (5) is challenging, especially under the single-input pinning constraint. Traditional analytical control methods usually rely heavily on accurate mathematical models and carefully designed controller structures. To address this issue, this paper adopts a data-driven DRL framework based on PPO. Before policy learning, a local node-selection analysis is first performed to provide theoretical guidance for choosing the pinning node. Then, the synchronization control problem is formulated as a continuous-state and continuous-action reinforcement learning task.

3.1. Local Pinning-Node Selection Analysis

To provide theoretical guidance for the selection of a single-input pinning node, this paper analyzes the local properties of the error system in the neighborhood of the synchronization manifold. The local control characteristics of Nodes 1, 2, and 3 are compared from three perspectives: local transverse stability, local linear quadratic regulation cost, and local control propagation capability. It should be emphasized that this analysis is used to establish a local ranking criterion for candidate pinning nodes, rather than to provide a strict global stability proof for the subsequent PPO-based closed-loop policy.

For the local pinning-node analysis, we first consider the baseline identical-system setting, where the drive and response systems share the same weight matrix, i.e., $W_r = W$, and no external disturbance is imposed. When the synchronization error approaches zero, i.e., $\mathbf{e} \rightarrow \mathbf{0}$, the nonlinear term can be expanded to the first order around the synchronization manifold. This gives the following local linear approximation of the error system [53]:

$$\dot{\mathbf{e}}(t) \approx [-I + W D(\mathbf{x}(t))] \mathbf{e}(t) + B u(t), \quad (6)$$

where I is the 3×3 identity matrix, W is the weight matrix of the drive system, $u(t) \in \mathbb{R}$ is the scalar control input, and $B \in \mathbb{R}^{3 \times 1}$ is the control allocation vector indicating the node at which the control signal is injected. The diagonal matrix $D(\mathbf{x})$ is composed of the derivatives of the activation function:

$$D(\mathbf{x}) = \text{diag} (1 - \tanh^2(x_1), 1 - \tanh^2(x_2), 1 - \tanh^2(x_3)), \quad (7)$$

where x_1, x_2 , and x_3 are the state variables of the drive system.

For single-input pinning control, different node selections correspond to different control allocation vectors:

$$B_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad B_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (8)$$

Here, B_1, B_2 , and B_3 represent the cases where the control input is injected into Nodes 1, 2, and 3, respectively.

First, to characterize the evolution tendency of small perturbations in the transverse subspace associated with different pinning nodes, the maximum conditional Lyapunov exponent (MCLE) [54] is introduced as a reference index for local transverse stability. Along the chaotic trajectory of the drive system, the instantaneous Jacobian matrix is evaluated, and a local variational analysis is performed in the transverse subspace complementary to the candidate pinning node. The corresponding MCLE is formally expressed as

$$\lambda_{\max}^{(c)}(B_i) = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{\|\delta \mathbf{e}_i(t)\|}{\|\delta \mathbf{e}_i(0)\|}, \quad i = 1, 2, 3, \quad (9)$$

where $\delta \mathbf{e}_i(t)$ denotes the perturbation trajectory in the transverse subspace complementary to the i th candidate pinning node, and $\|\cdot\|$ denotes the vector norm. If $\lambda_{\max}^{(c)}(B_i) < 0$, small perturbations in the corresponding transverse subspace tend to decay locally; if $\lambda_{\max}^{(c)}(B_i) > 0$, locally divergent directions still exist in that subspace. In this work, the MCLE is used as a local reference index for node pre-screening under the ideal pinning assumption.

Second, to compare the local control difficulty associated with different pinning nodes, a time-invariant local proxy model is further considered. Specifically, the Jacobian matrix is frozen at the reference point $\mathbf{x} = \mathbf{0}$, where $D(\mathbf{0}) = I$, leading to $A = W - I$. Based on this local approximation, the linear quadratic regulator (LQR) framework [55] is used here as an auxiliary indicator for comparing candidate pinning nodes. An LQR controller derived from the same local model is separately included as a model-based baseline in Section 4.

For each candidate input vector B_i , the standard LQR performance index is introduced as

$$J = \int_0^\infty (\mathbf{e}^T Q \mathbf{e} + u^T R u) dt, \quad (10)$$

where Q is the state weighting matrix and R is the input weighting matrix. The corresponding local state-feedback control law is

$$u = -K_i \mathbf{e}, \quad (11)$$

where the feedback gain matrix is given by

$$K_i = R^{-1} B_i^T P_i, \quad (12)$$

and P_i is determined by the continuous algebraic Riccati equation associated with (A, B_i) .

In this paper, the Riccati-based cost index is defined as

$$J_R(B_i) = \text{tr}(P_i), \quad (13)$$

which provides an initial-condition-independent relative measure of the local control difficulty associated with the candidate pinning node B_i . In addition, the gain norm $\|K_i\|$ is used to measure the required local feedback strength. These indices are used only for relative comparison among candidate nodes under the same local proxy model, rather than as the exact integral optimal cost for a specific initial condition.

Finally, to measure how effectively the control action propagates from different nodes to the entire local state space, the same frozen local matrix $A = W - I$ is used. Based on this matrix, the controllability matrices corresponding to the three candidate pinning nodes are constructed as [56]

$$C_i = [B_i, AB_i, A^2 B_i], \quad i = 1, 2, 3, \quad (14)$$

and the local control volume index is defined as

$$V_i = \det(C_i)^2. \quad (15)$$

Here, C_i characterizes the ability of the control input injected through the i th node to propagate over the entire local state space. Since this determinant-based index depends on the chosen state scaling, it is used only as a relative local propagation proxy among candidate nodes under the same coordinate system.

Table 1

Local auxiliary indicators for comparing candidate pinning nodes.

Pinning Node	$\lambda_{\max}^{(c)}$	J_R	$\ K_i\ $	V_i
Node 1	+0.6061	280.2686	20.3937	0.479
Node 2	+1.5793	15.5170	6.7286	262.18
Node 3	-0.5853	7.8327	4.3829	290.77

Table 1 shows that different pinning nodes exhibit significantly different local auxiliary indicators in the current continuous-time Hopfield chaotic neural network. For Node 3, $\lambda_{\max}^{(c)} < 0$, indicating that under the ideal pinning assumption, small perturbations in the transverse subspace complementary to this node tend to decay overall. Meanwhile, its J_R and $\|K_i\|$ are the smallest and V_i is the largest, suggesting relative advantages in both local control difficulty and control propagation capability. Therefore, Node 3 can be regarded as the preferred single-input pinning candidate for the current system.

By contrast, Node 2 has a relatively large V_i and a moderate J_R , implying that the control action can still propagate effectively over the local state space. However, its MCLE is positive, indicating that locally divergent directions still exist in the corresponding transverse subspace. Therefore, Node 2 is more suitable as a candidate node with certain

control potential but a more challenging synchronization task. Node 1 is clearly less favorable: its MCLE is positive, its J_R and $\|K_i\|$ are both significantly higher than those of the other two nodes, and its V_i is extremely small. This indicates weak local control efficiency and poor control propagation capability.

Based on the above theoretical indicators, the following local ranking can be obtained:

$$\text{Node 3} > \text{Node 2} \gg \text{Node 1}. \quad (16)$$

In summary, among the considered local auxiliary indicators, Node 3 shows the most favorable overall performance, followed by Node 2, whereas Node 1 performs worst. Therefore, for the current continuous-time Hopfield chaotic neural network, Node 3 is identified as the primary candidate for single-input pinning control, while Node 2 is retained as a challenging candidate for further evaluating the adaptability of the learned controller.

It should be emphasized that the MCLE $\lambda_{\max}^{(c)}$, Riccati-based cost index J_R , gain norm $\|K_i\|$, and local control volume index V_i reported in Table 1 are derived from the locally linearized error system near the synchronization manifold. These indicators provide local guidance for pinning-node selection, but they do not constitute a global stability guarantee for the PPO-based closed-loop policy. Therefore, the PPO experiments in Section 4 are used to empirically examine whether the local ranking can be reflected in actual nonlinear closed-loop synchronization.

3.2. PPO-Based Control Formulation and Training Procedure

After the pinning node is selected, the synchronization process of the drive–response system is formulated as a Markov decision process (MDP) with continuous states and continuous actions. In this formulation, the agent observes the current system information, outputs a bounded continuous action, and receives a reward determined by the synchronization error.

To enable the agent to perceive both the instantaneous dynamical state of the drive system and the current synchronization deviation, the full observation vector is defined as

$$O_t = \frac{1}{c_o} [x_1(t), x_2(t), x_3(t), e_1(t), e_2(t), e_3(t)]^T, \quad (17)$$

where $c_o = 5.0$ is a normalization factor used to suppress the influence of extreme state values on neural network training stability. The notation c_o is used here to avoid confusion with the Gaussian noise intensity σ used in the robustness experiments.

The policy network outputs a raw action a_t constrained to the continuous interval $[-1, 1]$. Considering the actuator magnitude limitation, it is mapped to the actual physical control input as

$$u(t) = \kappa a_t, \quad (18)$$

where κ is the control scaling factor, and $\kappa = 4$ is adopted in this paper.

In this study, the reward function is designed to mainly promote the rapid convergence of the synchronization error. Therefore, the reward is directly constructed from the global synchronization error:

$$r_t = -0.1 \|\mathbf{e}(t)\|_1 = -0.1 \sum_{i=1}^3 |e_i(t)|. \quad (19)$$

This reward has a clear physical interpretation: the smaller the synchronization error, the higher the reward. At the same time, it avoids introducing excessive manually designed shaping terms, thereby highlighting the direct response of policy learning to the synchronization objective itself. The trade-off between control effort and synchronization performance is further evaluated in the experimental section.

To learn a control policy in the above continuous-action setting, this paper employs the PPO algorithm [57]. PPO is a typical Actor–Critic method. By introducing a clipping mechanism to restrict the update magnitude of the policy, it achieves relatively high sample efficiency while limiting excessively large policy updates. The PPO objective function is written as

$$L(\theta) = \hat{\mathbb{E}}_t [\min (\rho_t(\theta) \hat{A}_t, \text{clip} (\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t)], \quad (20)$$

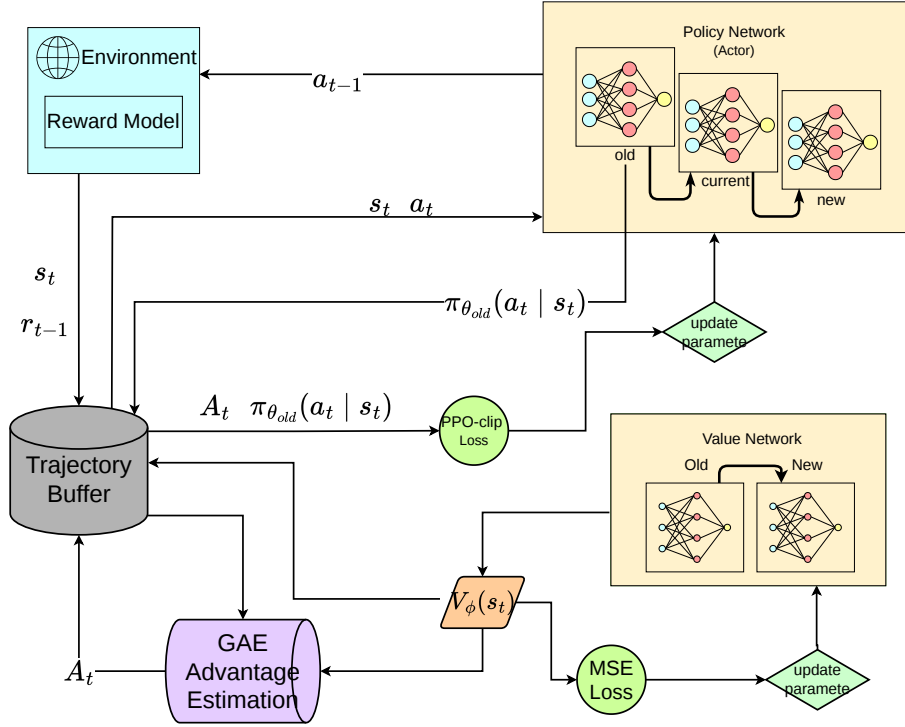


Figure 2: General training process of the PPO-based DRL framework.

where

$$\rho_t(\theta) = \frac{\pi_\theta(a_t|O_t)}{\pi_{\theta_{old}}(a_t|O_t)}. \quad (21)$$

where $\rho_t(\theta)$ denotes the likelihood ratio between the new and old policies, ϵ is the clipping hyperparameter, and $\hat{A}_t$ is the advantage estimate obtained using generalized advantage estimation (GAE). Through interaction with the continuous-time Hopfield chaotic system, PPO can progressively learn a nonlinear control policy that minimizes the long-term synchronization error under the constraint of single-input pinning control.

As shown in Fig. 2, PPO follows an interaction–update training process. The agent first interacts with the environment using the current policy and stores the collected states, actions, rewards, and value estimates in a trajectory buffer. The advantage function is then estimated, typically using GAE, and the policy network is updated by maximizing a clipped surrogate objective. Meanwhile, the value network is trained by minimizing the mean squared error between the predicted and target state values. Through repeated interaction and parameter updates, PPO improves the control policy while limiting excessively large policy changes during training.

The overall training procedure is summarized in Algorithm 1. In each iteration, trajectories are first collected by executing the current policy in the error-system environment. The collected data are then used to estimate the advantage function and update the policy network with the PPO clipped objective. The value network is updated by minimizing the squared value loss. After repeated interaction and optimization, the trained policy is used as the single-input pinning controller for the response system.

3.3. Boundedness and Post-Training Local Stability Assessment

Before evaluating synchronization performance, it is useful to distinguish three different properties: boundedness of the controlled trajectories, local stability of a deployed policy, and convergence of the PPO training algorithm. The first property can be established directly from the dissipative Hopfield dynamics and the bounded actuator. The second is assessed after training using the Jacobian of the actual closed-loop control-step map. No claim is made that PPO training itself guarantees global closed-loop stability.

Because the normalized action satisfies $a_t \in [-1, 1]$ and the physical action is defined by (18), the deployed control input satisfies

$$|u(t)| \leq \kappa, \quad \kappa = 4. \quad (22)$$

Moreover, each single-node pinning vector has $\|B_i\|_2 = 1$.

Proposition 3.1 (Boundedness of the controlled drive–response trajectories). *Consider the drive system (2) and response system (4), where W and W_r are finite constant matrices and the control input satisfies (22). For every finite initial condition, the corresponding trajectories are forward complete and uniformly ultimately bounded. In particular,*

$$\limsup_{t \rightarrow \infty} \|\mathbf{x}(t)\|_2 \leq \sqrt{3}\|W\|_2, \quad (23)$$

$$\limsup_{t \rightarrow \infty} \|\mathbf{y}(t)\|_2 \leq \sqrt{3}\|W_r\|_2 + \kappa. \quad (24)$$

Proof. Since $\|\tanh(\mathbf{z})\|_2 \leq \sqrt{3}$ for every $\mathbf{z} \in \mathbb{R}^3$, the Cauchy–Schwarz inequality and the induced matrix norm give

$$\begin{aligned} D^+ \|\mathbf{x}\|_2 &\leq -\|\mathbf{x}\|_2 + \sqrt{3}\|W\|_2, \\ D^+ \|\mathbf{y}\|_2 &\leq -\|\mathbf{y}\|_2 + \sqrt{3}\|W_r\|_2 + \|B_i\|_2 |u(t)| \\ &\leq -\|\mathbf{y}\|_2 + \sqrt{3}\|W_r\|_2 + \kappa, \end{aligned} \quad (25)$$

where D^+ is the upper right Dini derivative, $\|B_i\|_2 = 1$, and (22) has been used. Applying the scalar comparison lemma to (25) yields

$$\|\mathbf{x}(t)\|_2 \leq e^{-t} \|\mathbf{x}(0)\|_2 + \sqrt{3}\|W\|_2(1 - e^{-t}). \quad (26)$$

$$\|\mathbf{y}(t)\|_2 \leq e^{-t} \|\mathbf{y}(0)\|_2 + (\sqrt{3}\|W_r\|_2 + \kappa)(1 - e^{-t}). \quad (27)$$

Taking the upper limit as $t \rightarrow \infty$ gives (23) and (24). The explicit bounds also preclude finite escape; hence the locally Lipschitz state dynamics are forward complete. $\square$

For the nominal matched case $W_r = W$, the matrix in (3) has $\|W\|_2 = 8.9524$. Proposition 3.1 therefore gives the conservative bounds $\limsup_{t \rightarrow \infty} \|\mathbf{x}(t)\|_2 \leq 15.5059$ and $\limsup_{t \rightarrow \infty} \|\mathbf{y}(t)\|_2 \leq 19.5059$. These bounds are not intended to predict the observed attractor size. They establish non-divergence using worst-case norm inequalities. In particular, boundedness of both trajectories alone does not imply that the synchronization error converges to zero.

The MCLE $\lambda_{\max}^{(c)}(B_i)$ in Section 3.1 and the closed-loop exponent introduced below have different purposes. The former is a pre-training node-screening indicator derived from a locally linearized transverse model under the ideal pinning assumption and does not include the learned PPO policy. By contrast, the latter is evaluated after training from the actual deployed sample-and-hold closed-loop map. Consequently, their numerical values should not be directly compared or interpreted as equivalent stability measures.

The implementation uses a sample-and-hold controller: one action is evaluated every five RK4 substeps and is held constant over the control interval $\Delta t = 0.05$. Accordingly, local behavior of the trained controller is assessed using the actual discrete control-step map

$$\mathbf{x}_{k+1} = \Phi_x(\mathbf{x}_k), \quad \mathbf{e}_{k+1} = G_i(\mathbf{x}_k, \mathbf{e}_k; \pi_{\theta,i}), \quad (28)$$

where G_i includes the deterministic mean action of the trained policy, action clipping, the physical scale κ , and the five RK4 substeps. The transverse Jacobian of the deployed map is

$$M_{i,k} = \frac{\partial G_i}{\partial \mathbf{e}}(\mathbf{x}_k, \mathbf{e}_k; \pi_{\theta,i}). \quad (29)$$

The corresponding largest conditional Lyapunov exponent is numerically estimated as

$$\lambda_i^{\text{cl}} = \limsup_{N \rightarrow \infty} \frac{1}{N \Delta t} \ln \|M_{i,N-1} M_{i,N-2} \cdots M_{i,0}\|_2. \quad (30)$$

In practice, the Jacobians are obtained by automatic differentiation and the exponent is accumulated using QR renormalization to avoid numerical overflow. A negative estimate indicates that infinitesimal transverse perturbations around the evaluated closed-loop trajectory decay on average. This is a post-training, trajectory-dependent numerical assessment; it is not a proof of uniform contraction over the continuous state space, a global stability guarantee, or a convergence guarantee for PPO training. Parameter mismatch, impulse disturbance, and Gaussian noise are therefore treated as separate empirical robustness tests in Section 4, rather than as consequences of Proposition 3.1 or (30).

The numerical assessment was conducted for Nodes 2 and 3 using three independently trained policies for each node. Each policy was evaluated from ten initial conditions, giving 30 closed-loop trajectories per node and 60 trajectories in total. Each trajectory contained 2000 control intervals, and the first 100 intervals were discarded as a transient, leaving 95 time units for exponent accumulation. Table 2 summarizes the largest conditional Lyapunov exponent and the post-transient synchronization residual. The reported mean error is obtained by first averaging $\|\mathbf{e}_k\|_2$ over the post-transient portion of each trajectory and then averaging over the 30 trajectories for the corresponding node.

Table 2: Post-training local stability assessment of the deployed sample-and-hold PPO controllers.

Node	Runs	Mean λ_i^{cl}	Range	Mean $\ \mathbf{e}\ _2$
2	30	-1.117	$[-1.317, -0.872]$	0.0276
3	30	-0.581	$[-0.592, -0.557]$	0.0075

The largest conditional exponent is negative for all 60 evaluated trajectories. This consistently observed negativity provides numerical evidence that infinitesimal error perturbations decay on average near the tested closed-loop trajectories. A shorter check using 600 control intervals produced the same signs and similar aggregate values, supporting the numerical consistency of the finite-time estimates. Node 2 has a more negative mean exponent, whereas Node 3 has a smaller post-transient error. This difference is not contradictory: the conditional exponent measures the local average decay of an infinitesimal perturbation, while the residual error also depends on the location of the closed-loop trajectory and the learned control action. In particular, the policy architecture does not impose $\pi_{\theta,i}(\mathbf{x}, \mathbf{0}) = 0$ as an exact constraint. Therefore, the results support practical synchronization with small residual errors and post-training local average transverse stability, rather than exact asymptotic synchronization or a global stability certificate.

4. Experimental Results and Analysis

This section evaluates the proposed PPO-based single-input pinning synchronization strategy from three perspectives. First, the experimental configuration and evaluation metrics are specified. Second, the closed-loop performance under different pinning nodes and observation dimensions is examined to assess the practical relevance of the local node-ranking analysis in Section 3. Third, the learned policies are empirically evaluated under parameter mismatch [58], global impulse disturbance, and Gaussian noise perturbation.

4.1. Experimental Setup and Evaluation Metrics

The programs were implemented in Python using Stable Baselines3. The workstation is equipped with an NVIDIA RTX 5070Ti GPU; the policy training scripts considered here use the CPU. PPO was configured for 4×10^5 environment transitions, with four parallel environments and 512 steps per environment in each rollout. Completing the final rollout gives an actual checkpoint budget of 401,408 transitions.

During environment interaction, the continuous-time system dynamics were integrated using the classical fourth-order Runge–Kutta (RK4) method. The basic integration step size was set to 0.01 time units. To improve numerical stability, five RK4 substeps were performed within each control step. Therefore, each control update corresponded to an effective time interval of 0.05 time units. The maximum magnitude of the physical control input was constrained to 4 units.

The implementation also contains a numerical state guard at $[-10, 10]$ after RK4 substeps and disturbance updates. No guard activation was detected in the 2,940 reevaluated episodes covering the traditional baselines, matched-budget

Algorithm 1 PPO-based Single-Input Pinning Control for the Error System

Require: Error-system environment $\mathcal{E}$; policy network π_θ ; value network V_ϕ ; learning rate η ; clipping parameter ϵ ; discount factor γ ; GAE parameter λ .

Ensure: Optimized policy parameters θ .

- 1: Initialize policy parameters θ_0 and value-function parameters ϕ_0 .
- 2: Initialize the drive–response system and obtain the initial observation O_0 .
- 3: Define a trajectory buffer Γ to store observations, actions, rewards, and value estimates.
- 4: **for** $k = 0, 1, 2, \dots, K - 1$ **do**
- 5: Collect trajectories Γ_k by executing the current policy π_{θ_k} in the environment.
- 6: **for** each time step t in Γ_k **do**
- 7: Select action $a_t \sim \pi_{\theta_k}(\cdot|O_t)$.
- 8: Map the action to the physical control input $u(t) = \kappa a_t$.
- 9: Apply $u(t)$ to the response system and observe reward r_t and next observation O_{t+1} .
- 10: **end for**
- 11: Compute advantage estimates $\hat{A}_t$ using generalized advantage estimation.
- 12: Update policy parameters by maximizing the PPO clipped objective:

$$\theta_{k+1} = \arg \max_{\theta} \hat{\mathbb{E}}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip} \left(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right],$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(a_t|O_t)}{\pi_{\theta_k}(a_t|O_t)}.$$

- 13: Update value-function parameters by minimizing the squared value loss:

$$\phi_{k+1} = \arg \min_{\phi} \hat{\mathbb{E}}_t \left[\left(V_\phi(O_t) - \hat{R}_t \right)^2 \right].$$

- 14: **end for**

- 15: **return** optimized policy π_θ .

SAC comparison, and actuator-count ablation; the reported trajectories in these comparisons therefore evolve without state truncation.

At initialization, the initial states of both the drive system and the response system, denoted by $\mathbf{x}(0)$ and $\mathbf{y}(0)$, were sampled from the uniform distribution $[-1.0, 1.0]$. This interval covers the strongly nonlinear and non-saturated region of the $\tanh(\cdot)$ activation function, which increases the diversity of initial dynamical states and avoids ineffective simulations caused by excessive saturation.

For the reinforcement learning configuration, both the policy network and the value network were represented by multilayer perceptrons (MLPs). The Adam optimizer was adopted with a learning rate of 3×10^{-4} , so as to balance exploration efficiency and policy-update stability.

Three traditional controllers are included to provide references of increasing model dependence. First, the original fixed-gain feedback baseline [18] is retained without retuning:

$$u(t) = \begin{cases} -4, & -ke_i(t) < -4, \\ -ke_i(t), & -4 \leq -ke_i(t) \leq 4, \\ 4, & -ke_i(t) > 4, \end{cases} \quad (31)$$

where $e_i(t)$ denotes the synchronization error at the controlled node i . This baseline represents a simple local-error controller rather than an optimally tuned linear design. The previously selected gain $k = 8.0$ is used throughout all reported experiments.

Second, a continuous-time LQR baseline is obtained from the frozen local error model in Section 3. With $Q = I_3$ and $R = 1$, its saturated control law is

$$u_{\text{LQR},i}(t) = \text{sat}_4[-K_i \mathbf{e}(t)], \quad (32)$$

where $K_i = R^{-1} B_i^T P_i$ and P_i is the stabilizing solution of the continuous algebraic Riccati equation associated with (A, B_i) . The resulting gains are

$$\begin{aligned} K_2 &= (-0.116727, 3.871191, -5.502161), \\ K_3 &= (-0.171698, -2.085641, 3.850986). \end{aligned}$$

Unlike the fixed-gain baseline, LQR uses the complete synchronization-error vector and the nominal local model. The specified Q, R define a reproducible local design; optimality for the saturated nonlinear system is not claimed.

Third, a boundary-layer sliding-mode controller (SMC) [20, 21] is included as a nonlinear model-based baseline. Define

$$\mathbf{g}_0(\mathbf{x}, \mathbf{y}) = -\mathbf{e} + W [\tanh(\mathbf{y}) - \tanh(\mathbf{x})], \quad (33)$$

and select $c_i^T = K_i / (K_i B_i)$ so that $c_i^T B_i = 1$. The sliding variable and control law are

$$s_i = c_i^T \mathbf{e}, \quad u_{\text{SMC},i} = \text{sat}_4 \left[-c_i^T \mathbf{g}_0 - k_s \text{sat} \left(\frac{s_i}{\phi} \right) \right], \quad (34)$$

where $\text{sat}(z) = \max(-1, \min(1, z))$, $k_s = 16$, and $\phi = 0.5$. These parameters minimize the recorded validation score over $k_s \in \{4, 8, 12, 16\}$ and $\phi \in \{0.2, 0.5, 1, 2\}$, using ten validation seeds (0–9) separate from the test seeds. Validation includes nominal, mismatch ($\eta = 1$), and impulse scenarios; the score averages their MAEs normalized by 0.1, 0.15, and 0.45, respectively. Parameters are then frozen. Thus, although its compensation term uses the nominal model, SMC parameter selection has access to non-ideal validation scenarios. The saturated, sampled implementation is evaluated empirically; the continuous-time ideal sliding condition is not asserted to hold when the actuator saturates.

All four methods use the same RK4 solver, control-update interval, initial-state distribution, testing horizon (400 updates, or 20 time units), and actuator bound $|u| \leq 4$. The PPO results aggregate three independently trained policies (training seeds 42, 123, and 1024), each evaluated on 20 common test seeds from 100 to 119; each deterministic traditional controller is evaluated on the same 20 test seeds. For each scenario, the initial conditions and exogenous random realizations are shared across controllers. In the existing environment, drawing a mismatch matrix consumes random numbers before initialization; hence a fixed seed does not imply identical initial states between $\eta = 0$ and $\eta > 0$. Comparisons between controllers are paired within each scenario.

PPO is trained only on the nominal system. LQR uses $\mathbf{e}$ and the nominal local model, while SMC additionally uses $\mathbf{x}$ and $\mathbf{y}$ for nonlinear compensation. PPO receives $(\mathbf{x}, \mathbf{e})/5$ without evaluating an analytical model during deployment; its training simulator nevertheless uses the nominal dynamics. Fixed-gain feedback uses only e_i . Consequently, these are comparisons under matched actuation constraints, not identical information or design objectives. In particular, the PPO reward penalizes synchronization error but not control energy, whereas LQR explicitly penalizes both. Lower error at higher energy is therefore interpreted as a performance–effort tradeoff, not evidence of superior energy efficiency.

To evaluate the synchronization performance of different control methods, several metrics are introduced from the perspectives of overall synchronization accuracy, steady-state error, transient recovery capability, and control effort. Let $T = 400$ denote the number of control updates and $N = 3$ the system dimension. For the baseline and ablation comparisons, errors are recorded at the $T + 1$ sampling instants including initialization. The synchronization error vector is

$$\mathbf{e}(t) = \mathbf{y}(t) - \mathbf{x}(t) = [e_1(t), e_2(t), \dots, e_N(t)]^T. \quad (35)$$

The mean absolute error (MAE) is used to measure the average synchronization deviation over the whole testing interval:

$$\text{MAE} = \frac{1}{(T+1)N} \sum_{t=0}^T \sum_{i=1}^N |e_i(t)|. \quad (36)$$

MAE provides an intuitive measure of the overall error level by assigning equal weights to different time steps and state components.

The root mean square error (RMSE) is used to characterize the overall error level while being more sensitive to large deviations:

$$\text{RMSE} = \sqrt{\frac{1}{(T+1)N} \sum_{t=0}^T \sum_{i=1}^N e_i^2(t)}. \quad (37)$$

Compared with MAE, RMSE is more sensitive to transient large errors and is therefore useful for evaluating error fluctuations under parameter mismatch and impulse disturbance.

The final L1 error is used to measure the synchronization accuracy at the end of testing:

$$E_f = \|\mathbf{e}(T)\|_1 = \sum_{i=1}^N |e_i(T)|. \quad (38)$$

This metric reflects how closely the response system approaches the drive system at the final testing instant.

In the impulse disturbance experiment, the maximum post-pulse error is used to describe the largest error overshoot after the disturbance is injected:

$$M_p = \max_{t \geq t_p} \|\mathbf{e}(t)\|_1, \quad (39)$$

where t_p denotes the impulse injection step. The imposed state jump largely determines the immediate peak, so M_p alone is not interpreted as a controller-dependent recovery measure.

The post-pulse mean error is defined as the average global synchronization error from the impulse injection time to the end of testing:

$$E_{\text{post}} = \frac{1}{T - t_p + 1} \sum_{t=t_p}^T \|\mathbf{e}(t)\|_1, \quad (40)$$

where t_p is the impulse injection time and T is the final testing step.

The control energy is used to measure the cumulative control effort over the testing process, summing over all m active input channels:

$$E_u = \sum_{t=1}^T \sum_{j=1}^m u_j^2(t) \Delta t, \quad (41)$$

where $\Delta t = 0.05$ is the effective control update interval and $u_j(t)$ is the applied, saturated input during update t . The single-input comparisons have $m = 1$. The steady L1 error averages $\|\mathbf{e}(t)\|_1$ over samples $t = 320, \dots, 400$ (the final four time units), and differs from the terminal error E_f . Reported means first average the test initial conditions for each policy and then average policy seeds; the deterministic baselines have only the first level. Repeated initial conditions for the same trained policy are not treated as independent training replications.

In summary, MAE and RMSE are used to evaluate the overall synchronization performance, Final L1 Error is used to describe the terminal synchronization accuracy, M_p and E_{post} are used to characterize transient recovery capability under impulse disturbance, and E_u is used to assess the control cost.

4.2. Pinning-Node Selection and Observation Demand

To evaluate the guidance provided by the local node-selection analysis in Section 3, the single control input was first applied to Nodes 1, 2, and 3 under the same PPO framework and identical training configuration. The purpose of this experiment is not to recompute the local theoretical indicators, but to empirically examine how different pinning nodes affect synchronization reachability and training performance in the closed-loop data-driven control setting.

As shown in Fig. 3, the synchronization error evolves quite differently under different pinning nodes. When Node 3 is selected as the controlled node, the error decays more rapidly and remains at a relatively low level, indicating

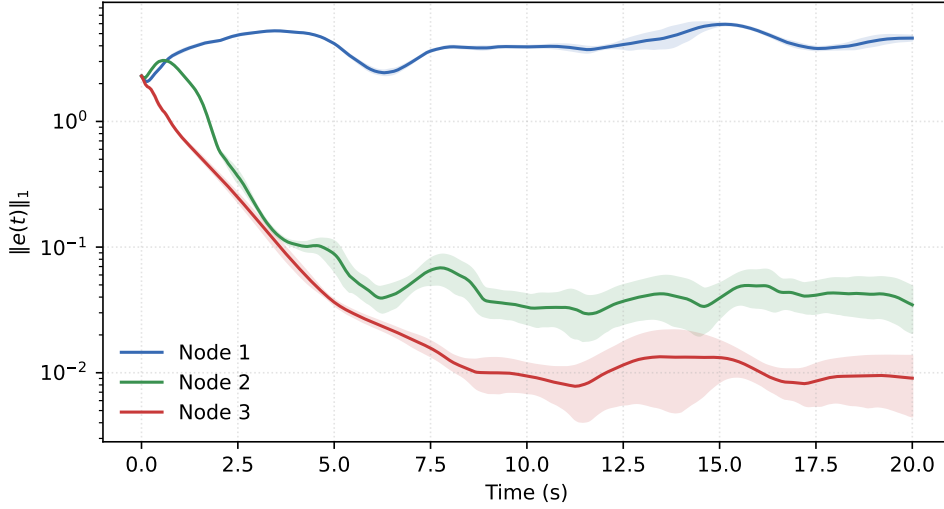


Figure 3: Synchronization error under different pinning nodes. Each curve averages three training-seed means, each evaluated on 20 common initial conditions (test seeds 100–119); shading shows one sample standard deviation across the three seed means.

the most favorable synchronization reachability. By contrast, although Node 2 can also achieve synchronization under certain conditions, its convergence speed and final error level are generally less stable than those of Node 3. Node 1 exhibits the weakest control performance and is more difficult to synchronize reliably under the same training setting.

To avoid relying only on a single error curve for qualitative judgment, MAE, RMSE, and Final L1 Error are further used to quantitatively compare the synchronization performance under different pinning nodes. The results in Table 3 further confirm the above observation. In terms of both average error and terminal error, Node 3 achieves the best overall synchronization performance. This result is generally consistent with the local auxiliary indicators in Table 1, including the $\lambda_{\max}^{(c)}$, J_R , $\|K_i\|$, and V_i . Therefore, the local theoretical analysis in Section 3 provides meaningful guidance for pinning-node selection in the considered Hopfield chaotic neural network.

Table 3

Quantitative synchronization performance under different pinning nodes, using three training seeds and 20 common test initial conditions per seed. MAE and RMSE are computed from the error components, consistently with the baseline and actuator-count comparisons.

Node	MAE	RMSE	Final L1 Error E_f
Node 1	1.3937	1.6342	4.6052
Node 2	0.0930	0.2863	0.0347
Node 3	0.0431	0.1434	0.0091

To further observe the state-component-level error evolution without the visual occlusion of three-dimensional surfaces, Fig. 4 presents two-dimensional heat maps obtained from the original multi-initial-condition trajectory data. Each row corresponds to a test trajectory, time is shown on the horizontal axis, and color represents the signed synchronization error. For Node 3, the error components e_1 , e_2 , and e_3 generally decay rapidly and remain close to zero after a short transient period. For Node 2, the error also shows a convergence tendency, but its transient fluctuation is more pronounced than that of Node 3. In contrast, for Node 1, large error fluctuations persist over a longer interval, indicating much weaker synchronization reachability. These observations are consistent with the quantitative results in Table 3 and further support the node ranking obtained in the local analysis.

Based on the above results, Node 3 is selected as the main pinning node in the subsequent experiments, since it exhibits the most favorable synchronization reachability in both local theoretical analysis and closed-loop PPO

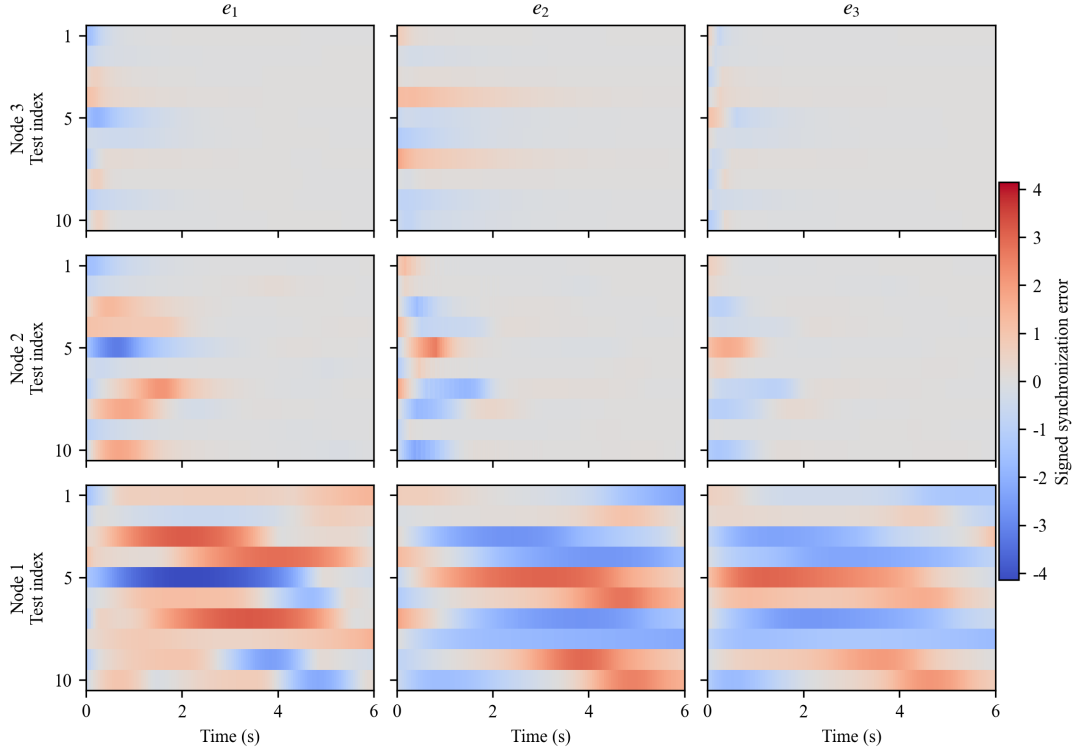


Figure 4: Two-dimensional heat maps of synchronization-error components under different random initial conditions. Rows correspond to Nodes 3, 2, and 1, columns correspond to e_1 , e_2 , and e_3 , and a common color scale is used throughout.

evaluation. Meanwhile, Node 2 is retained as a supplementary challenging scenario. Although it is not the locally preferred node, it still possesses certain control propagation capability and is therefore useful for testing the adaptability of the proposed method under a more difficult control channel.

4.2.1. Partial-observation assessment

Partial observability is next considered to examine how the information demand of the learned policy depends on the selected pinning node [36]. In practical control systems, complete state information may be unavailable because of sensor placement, communication bandwidth, or measurement limitations. The control structure, training procedure, and pinning node are fixed in this comparison, while only the observation supplied to the policy is changed.

For Node 3, the full-observation policy has access to all drive states and synchronization errors. The four-dimensional and two-dimensional observations are

$$O_{4D} = \{x_1, x_3, e_1, e_3\}, \quad (42)$$

and

$$O_{2D} = \{x_3, e_3\}, \quad (43)$$

respectively. For Node 2, the corresponding observations are $\{x_1, x_2, e_1, e_2\}$ and $\{x_2, e_2\}$.

Fig. 5 and Table 4 report the corresponding results. For Node 3, the full-observation and four-dimensional policies exhibit similar error evolution and both maintain a low later-stage error. Further compression to the two-dimensional local observation increases both MAE and Final L1 Error, indicating that excessive information reduction limits the policy's ability to infer the system dynamics.

The Node 2 results show a substantially larger degradation as observation information is removed. Thus, observation demand and pinning-node selection are coupled in this learning-based synchronization task: the locally favorable Node 3 remains effective with a reduced observation, whereas the more challenging Node 2 channel is more sensitive to information loss.

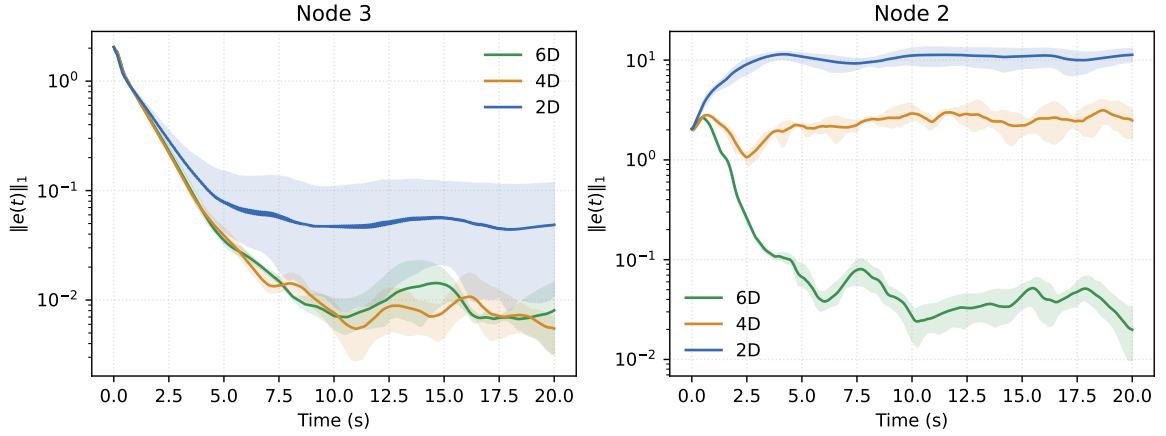


Figure 5: Comparison of L1 synchronization error under different observation configurations. Curves average three training-seed means (42, 123, and 1024), with ten common archived test initial conditions per seed (100–109); shading spans the minimum and maximum of the three seed means. This archived observation experiment uses fewer test initial conditions than the baseline and actuator-count comparisons.

Table 4

Synchronization performance under different observation settings, restricted to the same three training seeds and ten common test initial conditions. MAE is obtained by dividing the recorded time-averaged L1 error by $N = 3$. Observation components are specified in Eqs. 42–43.

Observation Setting	MAE	Final L1 Error
Node 3, 6D	0.0398	0.0081
Node 3, 4D	0.0390	0.0055
Node 3, 2D	0.0524	0.0487
Node 2, 6D	0.0761	0.0198
Node 2, 4D	0.7974	2.4874
Node 2, 2D	3.3271	11.3096

4.2.2. Actuator-count ablation

To quantify the performance cost of restricting actuation to one node, nominal PPO synchronization is evaluated for every nonempty subset of the three nodes: $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$, and $\{1, 2, 3\}$. For a subset S , the input matrix contains the corresponding columns of I_3 , and the policy produces $|S|$ actions. The six-dimensional observation, error-only reward, initial-state distribution, RK4 integration, policy hidden layers, PPO hyperparameters, and training budget of 401,408 transitions are unchanged. Each configuration includes seeds 42, 123, and 1024 and is evaluated on test seeds 100–119. All seven configurations and all three final checkpoints per configuration are reported, including configurations with poor synchronization.

The same per-actuator bound $|u_j| \leq 4$ is imposed in every configuration, while E_u sums energy across all active channels. Thus, additional actuators provide both additional control directions and a larger admissible total input magnitude ($\|u\|_2 \leq 4\sqrt{|S|}$). This is an actuator-availability comparison, not a comparison at fixed aggregate control authority. Only nominal synchronization is tested in this ablation. Actuator count measures the structural actuation requirement; hardware cost and implementation overhead are not measured.

Table 5 and Fig. 6 quantify the tradeoff. Single-node control at Node 3 gives MAE 0.0431 and energy 6.54, whereas $\{2, 3\}$ gives 0.0205 and 8.15 and $\{1, 2, 3\}$ gives 0.0118 and 10.91. Thus, one actuator retains low-residual synchronization with lower measured energy, but incurs a higher full-horizon error than these multi-input configurations. The steady L1 error does not decrease monotonically with actuator count: it is 0.0091 for $\{3\}$, compared with 0.0141 for

Table 5

Nominal PPO ablation over all nonempty actuator subsets. Each configuration uses three final policies and 20 common test initial conditions per policy. Values are mean $\pm$ SD across training-seed means; energy is summed over all active inputs.

Controlled nodes	Inputs	MAE	Steady L1 error	E_u
{1}	1	1.3937 ± 0.0029	4.2825 ± 0.1571	266.49 ± 13.41
{2}	1	0.0930 ± 0.0022	0.0426 ± 0.0131	9.09 ± 1.53
{3}	1	0.0431 ± 0.0012	0.0091 ± 0.0033	6.54 ± 0.34
{1, 2}	2	0.1145 ± 0.0110	0.0679 ± 0.0212	20.61 ± 1.28
{1, 3}	2	0.0232 ± 0.0007	0.0119 ± 0.0019	10.93 ± 0.12
{2, 3}	2	0.0205 ± 0.0011	0.0141 ± 0.0040	8.15 ± 0.27
{1, 2, 3}	3	0.0118 ± 0.0007	0.0105 ± 0.0026	10.91 ± 0.17

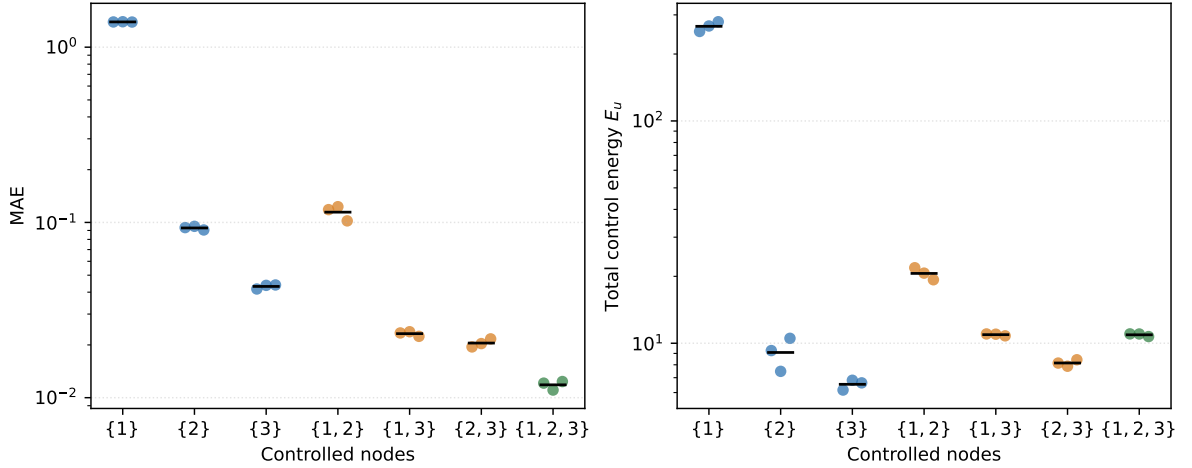


Figure 6: Nominal synchronization error and total control energy for all actuator subsets. Each colored point is a training-seed mean over 20 common test initial conditions; black horizontal marks denote the mean across three seeds. Blue, orange, and green indicate one, two, and three actuators, respectively.

{2, 3} and 0.0105 for {1, 2, 3}. The full-horizon improvement should therefore not be described as uniformly better steady-state precision.

The {1, 2} configuration has higher MAE and energy than {2} under the fixed training budget, while configurations containing Node 3 perform better in full-horizon MAE. This emphasizes the effect of actuator placement and finite-budget policy optimization. Since adding an actuator enlarges the admissible control set, the poorer learned result for {1, 2} is not evidence that the extra actuator intrinsically reduces controllability. These results support single-input control as a resource-constrained design choice, with an explicit performance cost, rather than as the best-performing architecture for every objective.

4.3. Robustness under Parameter Mismatch and External Perturbations

4.3.1. Parameter mismatch

After validating the pinning-node selection, the robustness of the learned control policy is first examined under parameter mismatch [42]. To model uncertainty in the response system, a mismatch is introduced into the weight matrix. Let W be the weight matrix of the drive system. The mismatched weight matrix of the response system is constructed as

$$W_r = W + \eta(W \odot \Delta), \quad (44)$$

where η denotes the mismatch intensity coefficient, $\odot$ denotes the Hadamard product, and $\Delta = [\delta_{ij}]$ is a random perturbation matrix with

$$\delta_{ij} \sim U(-0.03, 0.03). \quad (45)$$

This formulation corresponds to a relative multiplicative mismatch. That is, each nonzero connection weight in the response system is perturbed around its nominal value, while zero entries remain unchanged. This modeling approach preserves the original network topology and is also consistent with practical parameter deviations caused by manufacturing errors, calibration bias, and component tolerances.

It should be noted that η is a normalized scaling coefficient rather than the direct percentage of mismatch. When $\eta = 1.0$, the relative perturbation range is approximately $\pm 3\%$ of the original weight; when $\eta = 2.0$, the range is approximately $\pm 6\%$. In this work, the following mismatch levels are considered:

$$\eta \in \{0.0, 0.5, 1.0, 1.5, 2.0\}. \quad (46)$$

Here, $\eta = 0.0$ corresponds to the nominal case without mismatch, while larger values represent stronger parameter uncertainty.

All controllers are evaluated without adaptation under the perturbed response matrices. PPO is trained without domain randomization; LQR is designed from the nominal linearization, and SMC uses nominal compensation with parameters selected on separate validation scenarios as described above.

Fig. 7 reports the MAE and control energy of all four methods. Table 6 gives representative numerical results at the two endpoints of the mismatch range and also includes the impulse-response metric discussed below. For Node 2, the fixed-gain controller with the unchanged setting $k = 8$ cannot maintain a small error. PPO, LQR, and SMC are comparable in the nominal case; at $\eta = 2$, PPO attains an MAE of 0.1703, compared with 0.2330 for LQR and 0.2223 for SMC. The improvement in synchronization error is accompanied by higher control energy than these two model-based baselines. For Node 3, all methods except SMC remain relatively close in absolute MAE, and the traditional controllers generally require less energy. The benefit of PPO is therefore more evident for the less favorable Node 2 channel and under mismatch, rather than as a universal nominal-performance advantage.

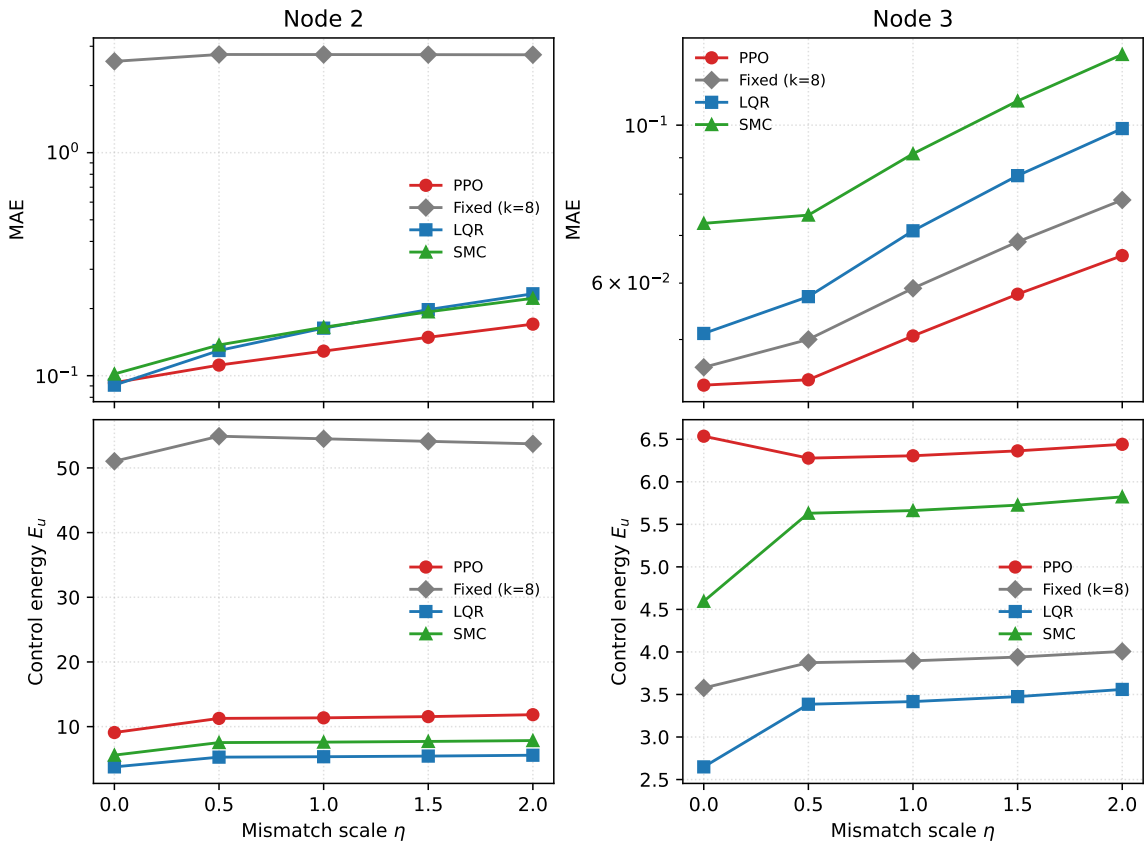


Figure 7: MAE and control energy of PPO, fixed-gain feedback ($k = 8$), LQR, and SMC under parameter mismatch. Curves show the mean over the common test initial conditions; PPO additionally aggregates three training seeds.

Table 6

Representative mismatch and impulse-disturbance results.

Node	Method	Nominal		$\eta = 2$	Impulse	
		MAE	E_u	MAE	E_{post}	E_u
2	PPO	0.0930	9.09	0.1703	2.0800	17.51
	Fixed ($k = 8$)	2.5673	51.03	2.7481	8.6188	74.02
	LQR	0.0907	3.78	0.2330	2.5264	9.95
	SMC	0.1018	5.59	0.2223	2.7691	14.13
3	PPO	0.0431	6.54	0.0656	1.2969	18.09
	Fixed ($k = 8$)	0.0457	3.58	0.0785	1.2518	16.21
	LQR	0.0510	2.65	0.0989	1.6883	7.49
	SMC	0.0728	4.60	0.1257	1.7352	11.37

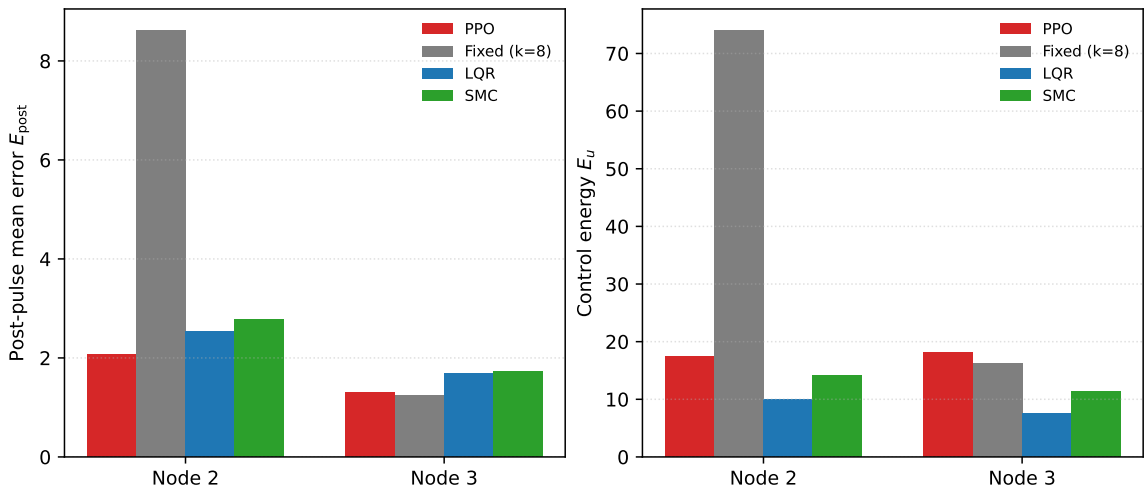
4.3.2. Impulse disturbance

In addition to parameter uncertainty, sudden impulse disturbance is also considered to evaluate the transient recovery capability of the controllers [43]. In the testing implementation, an impulse is added to all three drive-state components immediately after integration at step 200 ($t = 10$); the response state is not directly shifted. This tests recovery following an abrupt change of the synchronization reference. The impulse vector is set as

$$\mathbf{p} = [5, 5, 5]^T. \quad (47)$$

This setting avoids the asymmetry caused by disturbing only a single state component and provides a fairer comparison of recovery performance under different pinning nodes.

Because the same impulse is injected into all state components, the immediate peak M_p is dominated by the prescribed jump. The subsequent average E_{post} is therefore used as the main recovery metric. As shown in Fig. 8 and Table 6, PPO yields the lowest E_{post} for Node 2 (2.0800), improving over LQR (2.5264), SMC (2.7691), and fixed-gain feedback (8.6188). Its control energy is higher than that of LQR and SMC but substantially lower than that of the fixed-gain controller. For the favorable Node 3 channel, fixed-gain feedback and PPO give similar post-pulse errors (1.2518 and 1.2969, respectively), while LQR has the lowest energy. These results again indicate a channel-dependent tradeoff: PPO is most useful in the difficult Node 2 case, whereas a model-based controller is competitive when the local control channel is favorable.

**Figure 8:** Post-pulse mean error and control energy under the same global impulse disturbance for Nodes 2 and 3.

4.3.3. Gaussian-noise perturbation

To further evaluate the robustness of the learned controller under persistent random perturbations, zero-mean Gaussian noise is added to the response system during testing, while the training stage remains noise-free. This setting directly examines the generalization robustness of the trained policy under unseen stochastic disturbances. Specifically, after each control step and numerical integration, additive Gaussian noise is imposed on the response state:

$$\mathbf{y}_{k+1}^{\text{noisy}} = \mathbf{y}_{k+1} + \xi_k, \quad (48)$$

where

$$\xi_k \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_3). \quad (49)$$

Here, σ denotes the noise intensity and $\mathbf{I}_3$ is the 3×3 identity matrix. The tested noise levels are

$$\sigma \in \{0, 0.01, 0.02, 0.05\}. \quad (50)$$

The noise experiment is conducted for both pinning nodes. Fig. 9 shows that the MAE increases with σ for every controller. At $\sigma = 0.05$, PPO obtains the smallest mean MAE for both Node 2 and Node 3, as summarized in Table 7. In particular, for Node 2 its MAE is 0.2669, compared with 0.3742 for LQR and 0.3295 for SMC. This error reduction does not imply lower control cost: PPO uses considerably more energy than LQR, and SMC also becomes control-intensive under the state perturbations. Since noise is added to the evolving response state, this is a process-noise experiment, not a measurement-only noise test. For Node 3, fixed-gain feedback remains close to PPO in error while using much less energy. The noise results therefore indicate an empirical error–energy tradeoff within the tested range; they do not establish robustness for arbitrary noise intensities.

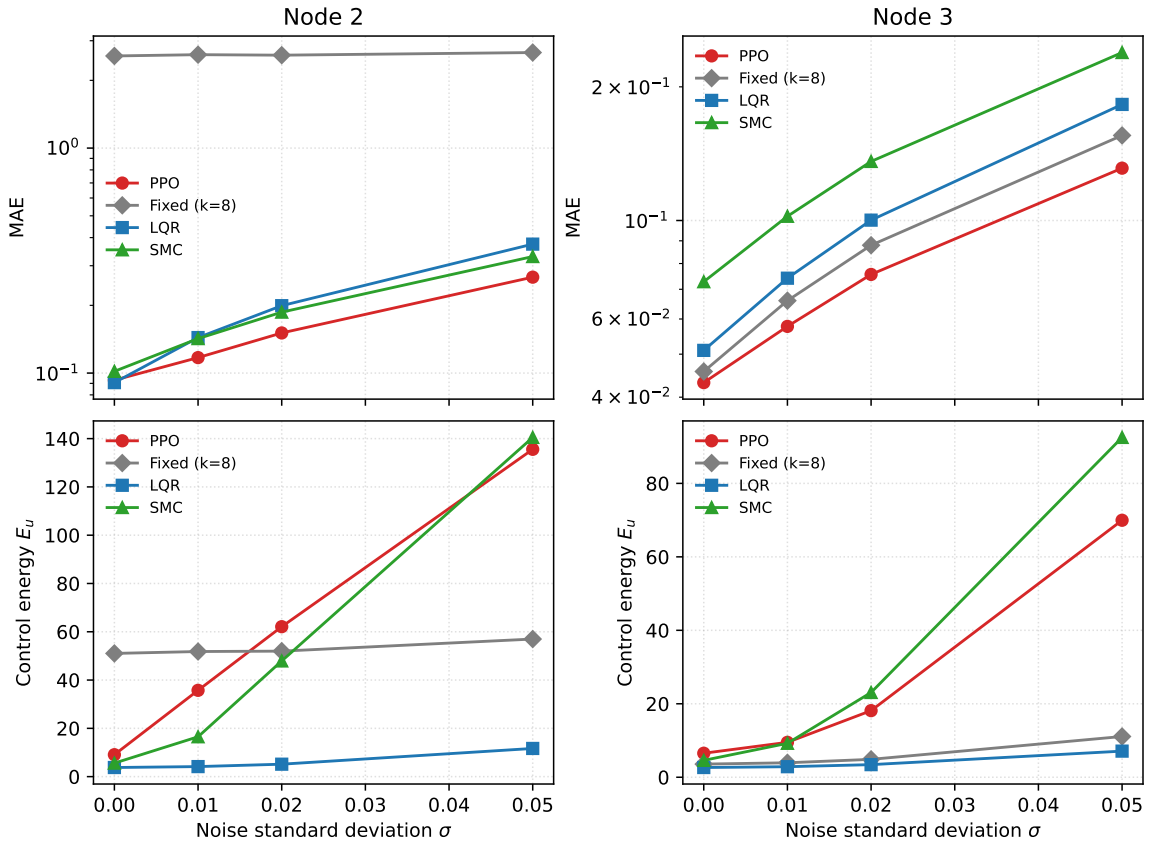


Figure 9: MAE and control energy of the four controllers under Gaussian-noise perturbation.

Table 7Performance at the highest tested noise intensity, $\sigma = 0.05$.

Node	Method	MAE	Steady L1 error	E_u
2	PPO	0.2669	0.5383	135.57
	Fixed ($k = 8$)	2.6584	8.7280	56.99
	LQR	0.3742	0.8979	11.67
	SMC	0.3295	0.7298	140.49
3	PPO	0.1312	0.3183	69.96
	Fixed ($k = 8$)	0.1555	0.4000	11.09
	LQR	0.1826	0.4419	7.11
	SMC	0.2389	0.6029	92.51

4.4. Comparison with Soft Actor-Critic

To assess the dependence of the results on the learning algorithm, soft actor-critic (SAC) [59] is evaluated on the same single-input Node 2 and Node 3 tasks. Both algorithms use the six-dimensional observation $(\mathbf{x}, \mathbf{e})/5$, reward $-0.1\|\mathbf{e}\|_1$, actuator bound $|u| \leq 4$, nominal training dynamics, and final checkpoints after 401,408 environment transitions. Three training seeds (42, 123, and 1024) and 20 common test initial conditions per seed are used. All checkpoints are retained, and evaluation uses deterministic actions without online learning. The compared test settings are nominal dynamics, mismatch $\eta = 2$, the drive-state impulse at $t = 10$, and response process noise with $\sigma = 0.05$.

Both implementations use two hidden layers of width 64, a learning rate of 3×10^{-4} , batch size 128, and discount factor 0.99. PPO uses Tanh hidden activations, four environments with 512 steps per rollout, ten optimization epochs, clip range 0.2, GAE parameter 0.95, and entropy coefficient 0.001. SAC uses ReLU hidden activations, one environment, a replay buffer of 10^5 transitions, 5,000 initial exploration steps, one gradient update per subsequent transition, target-update coefficient $\tau = 0.005$, and automatic entropy adjustment with target entropy -1 . These settings are fixed across nodes and seeds. Equal environment-transition budgets do not imply equal gradient-update counts or training times, and this comparison does not exhaust algorithm-specific hyperparameter tuning.

Table 8

PPO and SAC at the same budget of 401,408 environment transitions. Values are mean $\pm$ sample SD across three training-seed means, each obtained from 20 common test initial conditions. E_{post} is reported only for the impulse scenario.

Node	Scenario	Method	MAE	E_u	E_{post}
2	Nominal	PPO	0.0930 ± 0.0022	9.09 ± 1.53	—
2	Nominal	SAC	0.1173 ± 0.0046	6.27 ± 0.45	—
2	$\eta = 2$	PPO	0.1703 ± 0.0085	11.85 ± 1.60	—
2	$\eta = 2$	SAC	0.1513 ± 0.0037	9.31 ± 0.43	—
2	Impulse	PPO	0.4339 ± 0.0246	17.51 ± 1.19	2.0800 ± 0.1451
2	Impulse	SAC	0.4206 ± 0.0979	25.20 ± 9.79	1.9226 ± 0.5913
2	$\sigma = 0.05$	PPO	0.2669 ± 0.0049	135.57 ± 48.71	—
2	$\sigma = 0.05$	SAC	0.2632 ± 0.0052	86.13 ± 8.86	—
3	Nominal	PPO	0.0431 ± 0.0012	6.54 ± 0.34	—
3	Nominal	SAC	0.0416 ± 0.0010	4.60 ± 0.11	—
3	$\eta = 2$	PPO	0.0656 ± 0.0025	6.44 ± 0.30	—
3	$\eta = 2$	SAC	0.0644 ± 0.0009	5.07 ± 0.16	—
3	Impulse	PPO	0.2581 ± 0.0003	18.09 ± 3.50	1.2969 ± 0.0070
3	Impulse	SAC	0.2735 ± 0.0106	12.62 ± 0.53	1.3987 ± 0.0664
3	$\sigma = 0.05$	PPO	0.1312 ± 0.0066	69.96 ± 17.55	—
3	$\sigma = 0.05$	SAC	0.1273 ± 0.0013	50.40 ± 14.21	—

Table 8 and Fig. 10 show a scenario-dependent comparison. For Node 2, PPO has lower nominal MAE (0.0930 versus 0.1173), but SAC has lower MAE under mismatch $\eta = 2$ (0.1513 versus 0.1703) and also lower energy (9.31

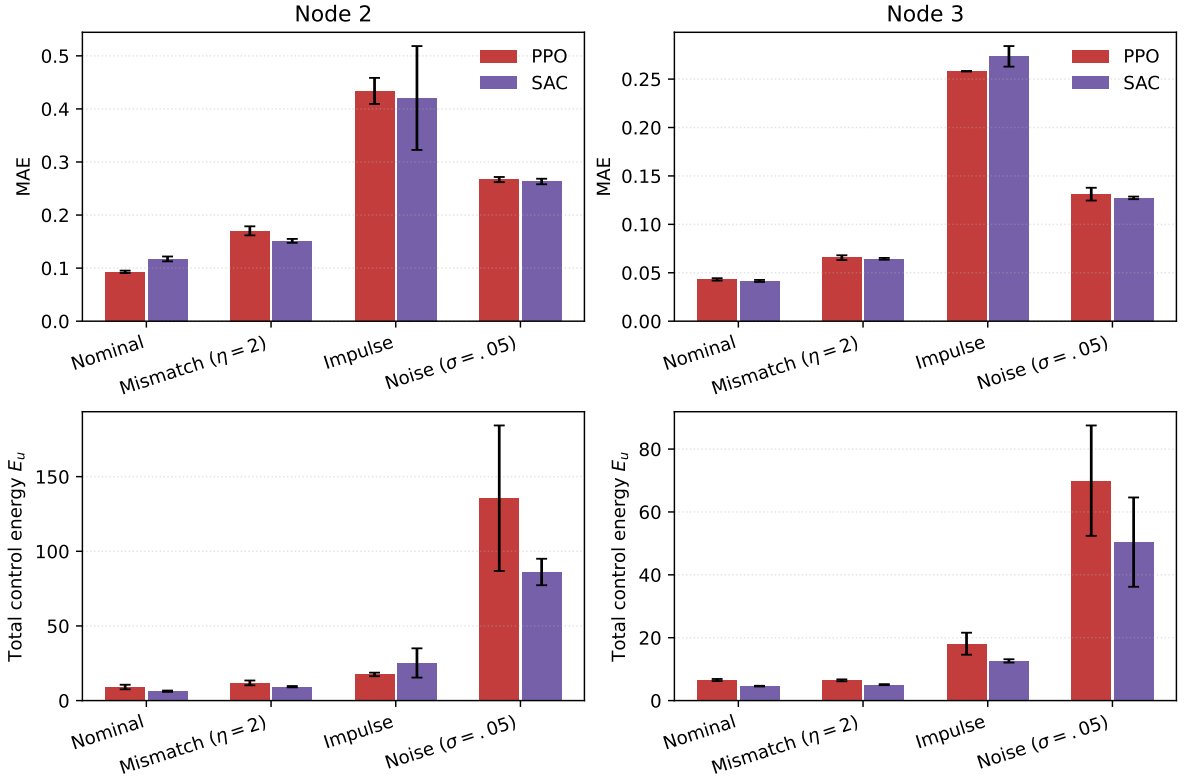


Figure 10: PPO–SAC comparison at matched environment-transition budgets. Bars show the mean of three training-seed means and error bars show their sample standard deviation. Each seed is evaluated on 20 common initial conditions. The upper panels use full-horizon MAE for every scenario; the separate post-impulse metric is given in Table 8.

versus 11.85). Following the impulse, SAC has lower mean E_{post} (1.9226 versus 2.0800), but greater variation across training-seed means (SD 0.5913 versus 0.1451) and higher mean energy (25.20 versus 17.51). At $\sigma = 0.05$, the mean MAEs are close (0.2632 for SAC and 0.2669 for PPO), while SAC uses less energy.

For Node 3, SAC has slightly lower mean MAE in the nominal, mismatch, and noise tests and lower mean energy in all four settings. PPO achieves lower post-impulse error (1.2969 versus 1.3987), at higher energy. Thus, the traditional-baseline advantages reported above do not imply that PPO outperforms another learning algorithm in every scenario. PPO is a viable implementation of the constrained pinning framework, while algorithm choice depends on the operating condition and the relative importance of error, variability, and control effort. With only three training seeds, the reported means and SDs are descriptive evidence rather than a general statistical superiority claim.

5. Image Encryption Scheme and Statistical Security Analysis

Astronomical observation systems generate large amounts of high-value image data, including celestial structures, planetary surfaces, and remote-sensing information. In distributed observation scenarios such as Very Long Baseline Interferometry [60], these images may need to be transmitted and shared across different observation sites, making them vulnerable to eavesdropping, tampering, and unauthorized access. Therefore, secure image protection is important for astronomical and space-related data applications.

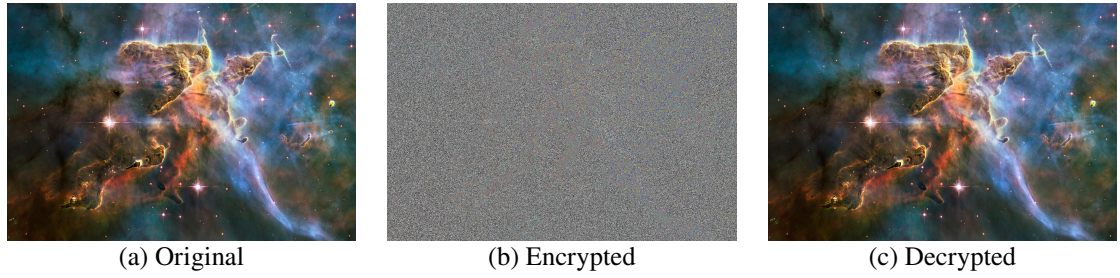


Figure 11: The original, encrypted, and decrypted images.

Astronomical images usually contain rich textures, complex intensity distributions, and strong spatial correlations among adjacent pixels, making them suitable test objects for image encryption algorithms. An effective encryption scheme should conceal the visual content of the original image and reduce its statistical redundancy. Since chaotic systems are sensitive to initial conditions and can generate pseudo-random trajectories, they have been widely used for image encryption. In a drive–response synchronization framework, synchronized chaotic trajectories can provide consistent key streams for encryption and decryption.

Motivated by this idea, this chapter applies the synchronized trajectories generated by the PPO-controlled continuous-time Hopfield chaotic neural network to image encryption. The flow chart of our method in image encryption is shown in Fig. 12. The proposed scheme combines synchronized Hopfield chaotic sequences, Logistic-map-based diffusion, and dynamic DNA coding. The plain image is first encrypted and then decrypted to verify reversibility. The RGB histogram and adjacent-pixel correlation are further analyzed to evaluate the statistical security of the encrypted image. The specific steps of encryption are as follows:

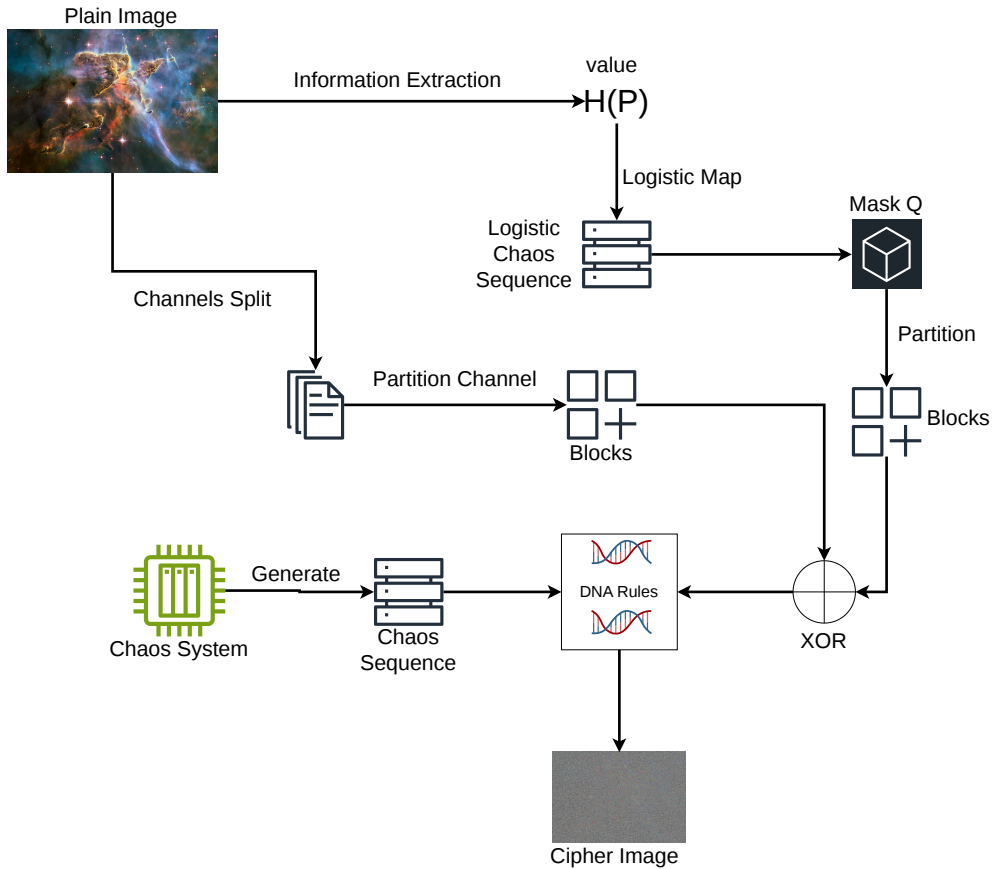


Figure 12: Flowchart of the image encryption process.

1. Let the plain image be defined as $P \in \mathbb{Z}_{256}^{H \times W \times C}$, where H , W , and C represent the height, width, and number of channels, respectively. The image is first split into individual channels:

$$P_c, \quad c \in \{1, 2, \dots, C\} \quad (51)$$

2. For each channel P_c , it is partitioned into $p \times p$ non-overlapping sub-blocks, with each sub-block having a size of $H_N \times W_N$, satisfying $H = pH_N$ and $W = pW_N$. This reorganization process can be rigorously defined by a tensor transformation $F : \mathbb{Z}^{H \times W} \rightarrow \mathbb{Z}^{p \times p \times D}$, where the output depth is $D = H_N W_N$. The transformation steps are as follows:

$$\mathbf{X} \xrightarrow{\text{Reshape}} \mathbf{X}_1 \in \mathbb{Z}^{p \times H_N \times p \times W_N} \quad (52)$$

$$\mathbf{X}_1 \xrightarrow{\text{Permute}(1,3,2,4)} \mathbf{X}_2 \in \mathbb{Z}^{p \times p \times H_N \times W_N} \quad (53)$$

$$\mathbf{X}_2 \xrightarrow{\text{Reshape}} \mathbf{Y} \in \mathbb{Z}^{p \times p \times D} \quad (54)$$

Applying this transformation to each channel yields:

$$B_c = F(P_c), \quad c = 1, 2, \dots, C \quad (55)$$

3. To extract the intrinsic features of the image, the information extractor $\mathcal{H}(P)$ is defined as:

$$\mathcal{H}(P) = \sum_{k=1}^C \sum_{i=1}^H \sum_{j=1}^W [(P(i, j, k) + 1) \times (i \cdot W + j + 1) \times k] \bmod M \quad (56)$$

where $P(i, j, k)$ denotes the pixel value at the corresponding 3D coordinates.

4. A dual-input hash mapping is designed to map the encrypted password $\mathcal{S}_{\text{password}}$ and the extracted image feature $\mathcal{H}(P)$ into an initial state vector z_0 :

$$z_0 = \text{Hash}(\mathcal{S}_{\text{password}}, \mathcal{H}(P)) \quad (57)$$

The vector z_0 serves as the initial state for the chaotic system. The Logistic map is employed as follows:

$$z_{t+1} = \mu z_t (1 - z_t) \quad (58)$$

By iterating the Logistic map with z_0 , a chaotic sequence vector z_L is generated. This sequence is then reshaped to construct the mask matrices Q and B_Q :

$$Q = \text{Reshape}_{H \times W} (\lfloor 10^{-4} \cdot z_L \rfloor \bmod 256) \quad (59)$$

$$B_Q = F(Q) \quad (60)$$

5. The mask matrix is utilized for the pre-diffusion of the image (where $\oplus$ denotes the bitwise XOR operation):

$$B_c = B_Q \oplus B_c, \quad c = 1, 2, \dots, C \quad (61)$$

To achieve comprehensive chained information interleaving across all channels and to resolve the undefined initial diffusion matrix, a circular XOR diffusion mechanism is designed. In the initial stage of the chained

Table 9
DNA coding rules

Rule	00	01	10	11
1	A	G	C	T
2	A	C	G	T
3	T	G	C	A
4	T	C	G	A
5	C	A	T	G
6	C	G	T	A
7	G	A	C	T
8	G	T	A	C

diffusion, the pre-diffused matrix of the last channel, B_C , is utilized as the initial feedback condition. It is XOR-ed with the first channel B_1 to establish a closed-loop diffusion ring:

$$D_1 = B_C \oplus B_1 \quad (62)$$

Subsequently, for the remaining channels, the sequential diffusion is strictly executed according to the forward chained rule:

$$D_c = D_{c-1} \oplus B_c, \quad c \in \{2, 3, \dots, C\} \quad (63)$$

Through this circular structure, any minute pixel perturbation in a single channel will be propagated to all channels during the subsequent diffusion and looping process, thereby effectively resisting differential attacks.

- Let R_i ($i = 1, 2, \dots, 8$) represent a set of encoding rules presents in Table 9 between numerical values and DNA sequences, with R_i^{-1} denoting the corresponding decoding process. For the state variable sequences x_1, x_2, x_3, x_4 outputted by the chaotic system, a quantization function F_{quantize} based on dead-zone characteristics is applied. This function quantizes the floating-point numbers into integers according to a maximum error bound ϵ :

$$s_i = F_{\text{quantize}}(x_i, \epsilon), \quad i \in \{1, 2, 3, 4\} \quad (64)$$

- Using these index sequences, DNA scrambling is executed. Two sequences are used for each round, resulting in two complete rounds of scrambling across the four sequences to yield the encrypted matrices:

$$D'_c = R_{s_2}^{-1}(R_{s_1}(D_c)) \quad (65)$$

$$D''_c = R_{s_4}^{-1}(R_{s_3}(D'_c)) \quad (66)$$

- Finally, the inverse tensor transformation is applied to the encrypted matrix to obtain the final encrypted image channels:

$$C_c = F^{-1}(D''_c) \quad (67)$$

The decryption process is the reverse of the encryption process. With the same password, the same Logistic mask, and the synchronized slave chaotic sequences, the DNA transformations and XOR diffusion can be inverted to recover the original image.

To illustrate the effectiveness of the proposed encryption method, we apply it to a color astronomical image. As shown in Fig. 11(a) and Fig. 11(b), the encrypted image exhibits a noise-like appearance and has no obvious visual similarity to the original image, indicating that the visual information is effectively concealed. Fig. 11(c) shows that the decrypted image is visually consistent with the original image, which verifies the reversibility of the proposed scheme when the correct password and synchronized chaotic key streams are used.

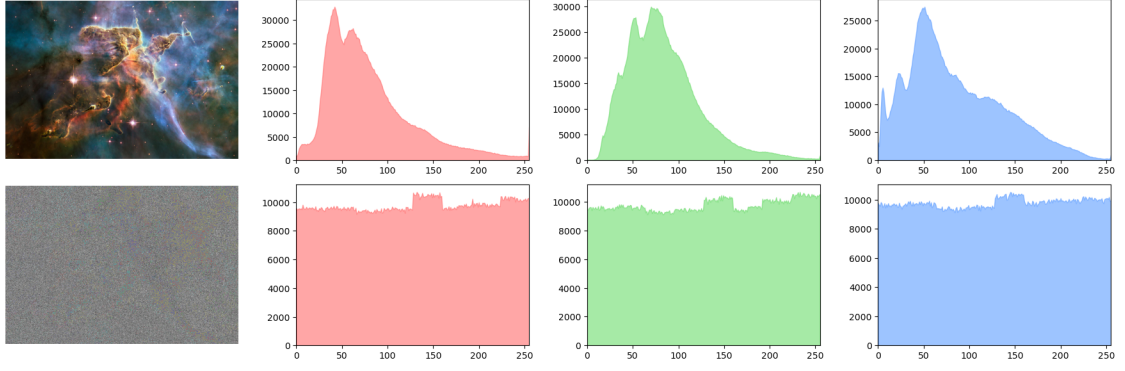


Figure 13: RGB histogram comparison between the plain image and the cipher image.

After visual inspection, histogram and adjacent-pixel correlation analyses are used to further evaluate the statistical properties of the cipher image. For each color channel $c \in \{R, G, B\}$, the histogram is defined as

$$H_c(k) = \#\{(i, j) \mid I_c(i, j) = k\}, \quad k = 0, 1, \dots, 255. \quad (68)$$

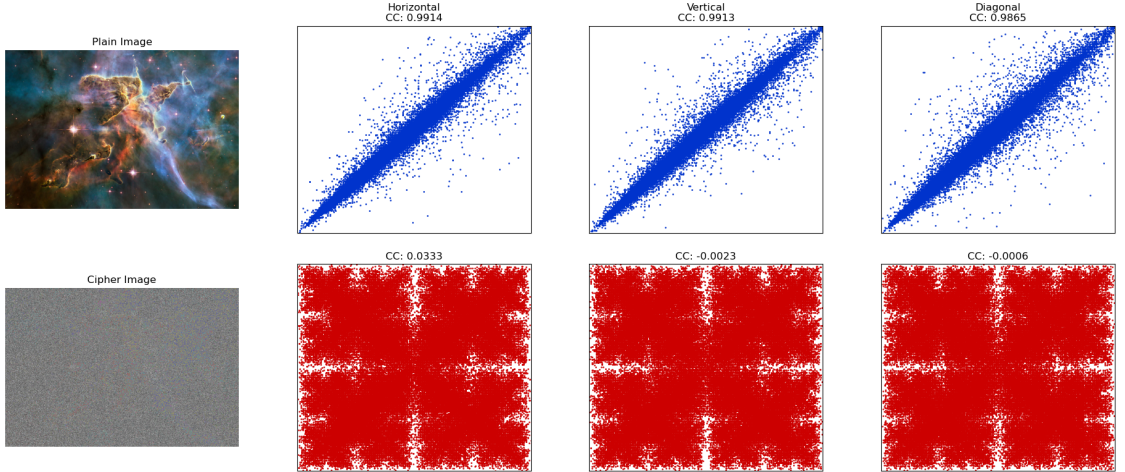


Figure 14: Adjacent-pixel correlation analysis of the plain image and the cipher image in horizontal, vertical, and diagonal directions.

As shown in Fig. 13, the histograms of the plain image are highly non-uniform, whereas those of the cipher image are much flatter over the gray-level range. This indicates that the encryption process weakens the original pixel-intensity distribution and reduces histogram-based statistical leakage.

The adjacent-pixel correlation coefficient is computed as

$$r_{xy} = \frac{\text{cov}(X, Y)}{\sqrt{D(X)}\sqrt{D(Y)}}, \quad (69)$$

where

$$\text{cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - E(X))(y_i - E(Y)), \quad (70)$$

and

$$D(X) = \frac{1}{N} \sum_{i=1}^N (x_i - E(X))^2. \quad (71)$$

Table 10

Correlation coefficients of the original and encrypted images

Image	Horizontal	Vertical	Diagonal
Original image of node2 model	0.9914	0.9913	0.9865
Encrypted image of node2 model	0.0333	-0.0023	-0.0006
Original image of node3 model	0.9921	0.9917	0.9852
Encrypted image of node3 model	0.0381	-0.0018	-0.0012
Encrypted image of Ref. [61]	-0.0015	0.0023	0.0021
Encrypted image of Ref. [62]	-0.0123	-0.0068	-0.0368

Here, X and Y denote two adjacent-pixel sequences sampled in the horizontal, vertical, or diagonal direction. As reported in Table 10, the original images have correlation coefficients close to 1 in all three directions, confirming the strong spatial redundancy of natural images. After encryption, the coefficients produced by both the node2 and node3 models decrease to values close to 0. The scatter plots in Fig. 14 also show that the encrypted pixels are broadly dispersed instead of being concentrated along the diagonal line. These results indicate that the proposed encryption process effectively suppresses adjacent-pixel correlation.

It should be noted that the node1 model is not included because its synchronization performance is insufficient, causing the generated master-slave key streams to fail the sequence consistency check required for correct encryption and decryption. Both node2 and node3 are evaluated under the six-dimensional full-observation setting. Compared with the encrypted images reported in Refs. [61, 62], the proposed method achieves comparable correlation suppression, demonstrating its effectiveness in reducing local statistical redundancy.

6. Conclusion

This paper has investigated the single-input pinning synchronization problem for two continuous-time Hopfield chaotic neural networks and developed a PPO-based data-driven control framework. In the considered setting, only one scalar control input is injected into a selected node of the response system. A local pinning-node analysis was conducted by combining several complementary indicators, and the obtained node-ranking result was found to be generally consistent with the closed-loop synchronization behavior. For the deployed controller, state boundedness under the actuator constraint was established analytically, while negative largest conditional Lyapunov exponents across the evaluated trajectories provided post-training local numerical evidence of average transverse error decay. These results support practical low-residual synchronization rather than a global or exact asymptotic stability guarantee. The empirical tests further show that the PPO-implemented controller can maintain effective synchronization under nominal and non-ideal settings, including parameter mismatch, impulse disturbance, Gaussian noise perturbation, and partial observation. Comparisons with the unchanged fixed-gain controller ($k = 8$), LQR, and sliding-mode control show that PPO has its clearest error-recovery advantage for the less favorable Node 2 channel under mismatch and external perturbations. The model-based baselines remain competitive in the nominal Node 3 case and often use less energy, revealing that the advantage is robustness- and channel-dependent rather than universal.

The matched-budget SAC comparison further shows that the observed advantages are algorithm- and scenario-dependent: SAC is competitive and improves several error or energy metrics, while PPO retains advantages in nominal Node 2 error and Node 3 impulse recovery. The all-subset actuator ablation shows that single-node control can reduce actuator count and measured control energy, with a cost in full-horizon synchronization error relative to favorable multi-input configurations. These comparisons support a constrained-control design tradeoff rather than universal superiority of PPO or of single-input actuation.

The synchronized Hopfield chaotic trajectories were further applied to an image encryption and decryption task through Logistic-map diffusion and dynamic DNA coding. The experimental results show that the synchronized trajectories can provide consistent key streams for encryption and decryption, while the encrypted images present noise-like visual appearance and reduced statistical redundancy. These findings suggest that DRL-based single-input pinning control is not only feasible for constrained synchronization of Hopfield chaotic neural networks, but also has potential application value in chaos-based secure image protection. Future work will focus on strengthening theoretical stability analysis, extending the framework to larger-scale or hardware-implemented chaotic neural networks, and conducting more comprehensive security evaluation for the encryption application.

Declaration of competing interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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